1 Data Set:

Outlook	Temperature	Humidity	Windy	Rain
O	T	H	W	R
Sunny	Hot	high	weak	NO.
Sunny	Mild	high	Strong	NO
Cloudy	Hot	high	weak	Yes
Rainy	Mild	high	Weak	Yes
Rainy	Cool	Normal	weak	Yes
Rainy	Cool	Normal	Strong	NO
Cloudy	Mild	Normal	Strong	Yes.
Sunny	Cool	Normal	Weak	NO

Step 1:

Calculate the entropy of the target Rain:

Total = 8

- Yes = 4 } binary
- NO = 4 }

2

Formula:

$$-(\text{Yes}\log\text{Yes} + \text{No}\log\text{log})$$

$$-\frac{4}{8}\log\frac{4}{8} + -\frac{4}{8}\log\frac{4}{8}$$

$$-0.5\log 0.5 + -0.5\log 0.5$$

$$-0.5 \times \frac{\log(0.5)}{\log(2)} + -0.5 \times \frac{\log(0.5)}{\log(2)}$$

$$-0.5 \times -1 + -0.5 \times -1$$

$$0.5 + 0.5$$

Ram Ed = 1

Step 2:

Calculate Information Gain

A. Information gain of outlook

outlook	Yes	No	Entropy
• Sunny	0	3	0
• cloudy	2	0	0
• Rainy	2	1	0.917

⇒ Calculate of each:

- Rainy:
  - Yes = 2
    No = 1
    Total = 3

- $-\frac{2}{3}\log\frac{2}{3} + -\frac{1}{3}\log\frac{1}{3}$

$-0.667 \log 0.6667 + -0.333 \log 0.333$

$-0.667 \frac{\log(0.667)}{\log(2)} + -0.333 \frac{\log(0.333)}{\log(2)}$

$-0.667 \times -0.584 + -0.333 \times -1.586$

$0.389 + 0.528$

[Rainy Entr 0.917]

⇒ Calculate Weight Entropy

$\frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(0.917)$

[outlook = 0.343]

c) IG of outlook

$1 - 0.343$

B= Information Gain of Temp

Temp	Yes	No	Entropy
Hot	1	1	1
Mild	2	1	0.917
Cool	1	2	1

Calculate of each

→ Hot

Yes = 1

No = 1

Total = 2

$$-\frac{1}{2}\log\frac{1}{2} + -\frac{1}{2}\log\frac{1}{2}$$

$$-0.5 \times \frac{\log(0.5)}{\log(2)} + 0.5 \times \frac{\log(0.5)}{\log(2)}$$

$$-0.5 \times -1 + 0.5 \times -1$$

$$0.5 \quad 0.5$$

[Hot = 1]

Mild

Yes =

NO =

Total

$= \frac{2}{3}$ log

$-0.667 \log$

-0.667

Cool

Ye

No

To

⇒ Mild

Yes = 2

NO = 1

Total = 3

$$-\frac{2}{3}\log\frac{2}{3} + -\frac{1}{3}\log\frac{1}{3}$$

$$-0.667\frac{\log(0.667)}{\log(2)} + -0.333\frac{\log(0.333)}{\log(2)}$$

$$-0.667 \times -0.584 + -0.333 \times -1.586$$

$$0.384 + 0.528$$

Mild = 0.917

Cool

Yes = 1

NO = 2

Total 3

$$-\frac{1}{2}\log\frac{1}{2} + \frac{-1}{2}\log\frac{1}{2}$$

$$-0.5 * \frac{\log(0.5)}{\log(2)} + -0.5\frac{\log(0.5)}{\log(2)}$$

$$0.5 + 0.5$$

Cool = 1

=> weigh Entropy

$$\frac{2}{8}(1) + \frac{3}{8}(0.917) + \frac{3}{8}(1)$$

[Temp = 0.968]

=> IG of Temp

$$1 - 0.968$$

[IG temp = 0.032]

C = Information Gain of Humidity

Humidity	Yes	NO	Entropy
high	2	2	1
Normal	2	2	1

Calculat for each

high:

Yes = 2

NO = 2

Total = 4

$$-\frac{2}{4}\log\frac{2}{}$$

$$-0.5\log($$

$$-0.5\left(\frac{\log}{\log}\right.$$

$$-0.5\ V$$

$$0.5$$

[h

Normal

Yes =

NO =

Total

$$-\frac{2}{4}$$

7

$$-\frac{2}{4}\log\frac{2}{4} + -\frac{2}{4}\log\frac{2}{4}$$

$$-0.5\log(0.5) + (-0.5\log 0.5)$$

$$-0.5\left(\frac{\log(0.5)}{\log(2)}\right) + -0.5\frac{\log(0.5)}{\log(2)}$$

-0.5×-1 $\quad +0.5 \times -1$

0.5 $\quad 0.5$

[high = 1]

=) Normal

Yes = 2

No = 2

Total = 4

$$-\frac{2}{4}\cdot\log\left(\frac{2}{4}\right) + -\frac{2}{4}\log\left(\frac{2}{4}\right)$$

0.5 $+$ 0.5

[Normal = 1]

weight Entropy

$\frac{4}{8}(1) + \frac{4}{8}(1)$

$0.6 + 0.5$

$\log_2(1)$

IG of humidity

$1 - 1$

0

Windy	Information Gain		Entropy
Windy	Yes	NO	0.9
Weak	3	2	
Strong	1	2	

Calculate of each

weak

Yes = 3

NO = 2

Total = 5.

$\frac{-3}{5}$ log

-0.6

-0.6

Strong

Yes =

NO

Tota

1

3

eigh

9

$$\frac{-3}{5}\log\frac{3}{5} + \frac{-2}{5}\log\frac{2}{5}$$

$$-0.6\,\frac{\log(0.6)}{\log(2)} + -0.4\,\frac{\log(0.4}{\log(2)}$$

$$-0.6 \times -0.736 + 0.441 + 0.528$$

$$\boxed{0.97}$$

Story

Yes = 1

NO = 2

Total = 3

$$-\frac{1}{3}\log\frac{1}{3} + -\frac{2}{3}\log\frac{2}{3}$$

$$\boxed{0.918}$$

Weigh of Windy

$$\frac{5}{8}(0.97) + \frac{3}{8}(0.918)$$

$$\boxed{0.950}$$

If of wind
1 - 0.950
If wind 0.05

Rain: 1
outlook 0.657
Temp: 0.932
Humidity: 0
Wind = 0.05

The root Node = 0.65
outlook